

FALLACIES

Computer Science — First Semester | Lecture 6

Learning objective: Understand what fallacies are, recognize common errors in reasoning, and apply critical thinking when evaluating everyday and technical arguments.

6.1 What is a fallacy?

A **fallacy** is an error in reasoning that weakens or undermines an argument. A fallacy can make an argument sound convincing even though its reasoning is flawed.

Fallacies occur in everyday conversations, advertisements, academic discussions, debates, and technical decision-making. A fallacious argument can sometimes reach a true conclusion. The problem lies in the reasoning, not necessarily in the final conclusion.

6.2 Main types of fallacies

Fallacies are commonly divided into two broad categories:

Formal fallacies	Informal fallacies
Errors in the logical structure of an argument. They can often be identified by examining the argument's form.	Errors involving relevance, assumptions, evidence, ambiguity, or the content of an argument.

A. Formal fallacies

A formal fallacy occurs when an argument has an invalid logical structure.

Example: Affirming the consequent

- If it rains, the road becomes wet.
- The road is wet.
- Therefore, it rained.

This reasoning is invalid because the road might be wet for another reason, such as someone washing it.

Symbolic form: $P \rightarrow Q$; Q ; therefore P . This is the fallacy of affirming the consequent.

The valid form called *modus ponens* is: If P then Q ; P ; therefore Q .

B. Informal fallacies

Informal fallacies involve problems with the content or relevance of reasoning. They cannot always be detected by looking only at the argument's symbolic structure.

6.3 Common informal fallacies

1. Ad hominem (attacking the person)

Instead of addressing an argument, someone attacks the person presenting it.

Example: “Your suggestion about improving our university's computer lab is wrong because you are only a first-semester student.”

Why it is fallacious: The speaker attacks the student's experience rather than examining the suggestion itself.

2. Bandwagon fallacy (appeal to popularity)

Assuming that something is correct or superior merely because many people believe or use it.

Example: “Millions of people use this software, so it must be the most secure software.”

Why it is fallacious: Popularity alone does not establish security.

3. Straw man fallacy

Misrepresenting someone's argument to make it easier to attack.

Example: Student A says, “We should reduce unnecessary screen time during lectures.” Student B replies, “So you want us to stop using computers completely?”

Why it is fallacious: Student B exaggerates the original proposal rather than responding to it accurately.

4. Slippery slope fallacy

Assuming that one event will inevitably trigger a chain of increasingly serious consequences without sufficient evidence.

Example: “If we allow students to submit one assignment late, eventually nobody will submit assignments on time.”

Why it is fallacious: The predicted chain is asserted without adequate support. A chain of consequences can be reasonable when supported by evidence.

5. Appeal to authority

Treating a claim as proven solely because an authority or famous person said it.

Example: “A famous actor says this programming technique is the fastest, so it must be the fastest.”

Why it is fallacious: Fame does not establish technical expertise or provide benchmark evidence. Qualified expert testimony can still be useful evidence.

6. Hasty generalization

Drawing a broad conclusion from too few or unrepresentative examples.

Example: “I encountered two bugs in Python programs, so Python is an unreliable language.”

Why it is fallacious: Two examples are insufficient to establish the general reliability of an entire programming language.

7. False dilemma (false dichotomy)

Presenting only two possible choices when more alternatives exist.

Example: “Either you study computer science, or you will never have a successful career.”

Why it is fallacious: Many other educational and career paths can lead to success.

8. Circular reasoning

Using the conclusion as part of the evidence supporting that same conclusion.

Example: “This program is trustworthy because it is a reliable program.”

Why it is fallacious: The statement repeats the claim without providing independent evidence.

6.4 Fallacies in computer science

Fallacies can affect technical decisions as well as everyday arguments. Here are common examples:

Situation	Fallacious reasoning	Problem
Choosing software	“This software is popular, so it must be secure.”	Popularity does not prove security.
Debugging	“The program worked yesterday, so today's error must be caused by the compiler.”	Ignores other possible causes.
Evaluating performance	“My program ran faster once, so it is always faster.”	One test does not establish consistent performance.
Choosing a database	“My friend uses this database, so it must be suitable for our project.”	Personal preference does not establish project suitability.

In computer science, good reasoning means testing assumptions, collecting evidence, comparing alternatives, and verifying results rather than relying on unsupported claims.

6.5 How to identify a fallacy

When you encounter an argument, ask these five questions:

1. What is the main conclusion?
2. What premises are being used to support it?
3. Are the premises relevant to the conclusion?
4. Is there enough evidence to justify the conclusion?
5. Does the reasoning contain a structural error, unsupported assumption, or misleading claim?

These questions help distinguish sound reasoning from arguments that only appear convincing.

6.6 Practice quiz

1. A student says: “My programming teacher is wrong because he is old.” Which fallacy is this?
A. Bandwagon
B. Ad hominem
C. Hasty generalization
D. False dilemma
2. “Every time I use this app, it crashes. Therefore, all apps made by this company are unreliable.” This is:
A. Circular reasoning
B. Slippery slope
C. Hasty generalization

D. Modus ponens

3. If P implies Q, and Q is true, therefore P is true. This is:

- A. Modus ponens
- B. Affirming the consequent
- C. Modus tollens
- D. A sound argument

4. "Either you learn Java, or you will never become a programmer." Which fallacy is this?

- A. Straw man
- B. Appeal to authority
- C. False dilemma
- D. Circular reasoning

5. A famous singer recommends a laptop, so it must be the best laptop for programming. This is:

- A. Appeal to authority
- B. Hasty generalization
- C. Modus ponens
- D. Valid deduction

Answer key with explanations

Answer: B — Ad hominem. The student attacks the person rather than addressing the teacher's argument.

Answer: C — Hasty generalization. A broad conclusion is drawn from limited personal experience.

Answer: B — Affirming the consequent. Q may be true for reasons other than P.

Answer: C — False dilemma. It presents only two options while ignoring other programming languages and learning paths.

Answer: A — Appeal to authority. The singer's fame is not sufficient evidence of expertise in evaluating programming laptops.

Lecture summary

- A fallacy is an error in reasoning.
- Formal fallacies involve invalid logical structures.
- Informal fallacies involve problems such as irrelevant evidence, misleading language, and unsupported assumptions.
- Common fallacies include ad hominem, bandwagon, straw man, slippery slope, hasty generalization, false dilemma, appeal to authority, and circular reasoning.
- Identifying fallacies helps computer scientists make better technical decisions and evaluate arguments critically.